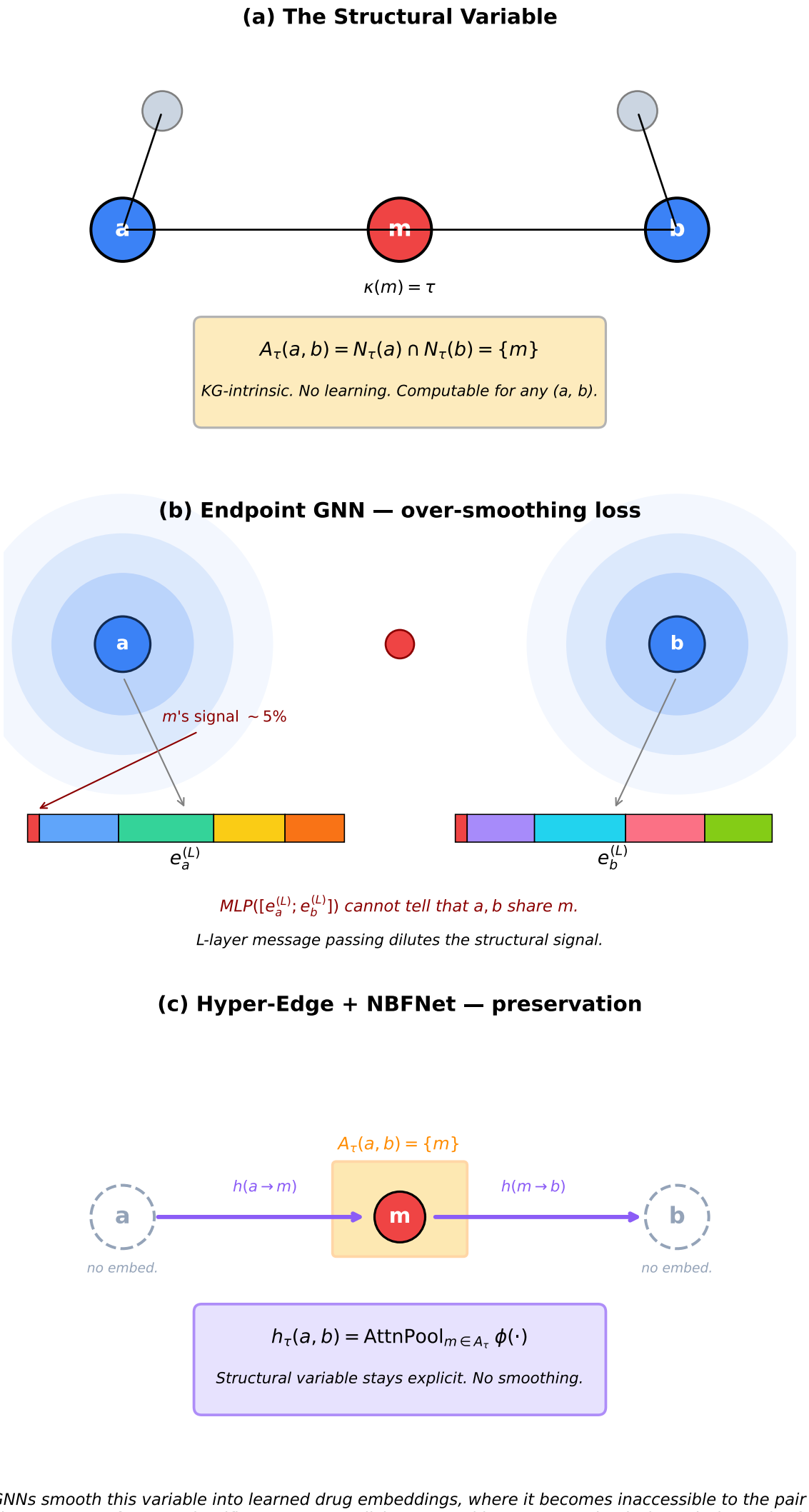
Figure 1. Cold-start DDI requires preserving the KG structural variable $A_\tau(a, b)$ — the typed middle common neighbors of the drug pair.



Endpoint GNNs smooth this variable into learned drug embeddings, where it becomes inaccessible to the pair classifier. Our hyper-edge + NBFNet framework reifies A_τ as an explicit structural input, preserving it through the entire forward pass.